



Head banging

Every time a woodpecker's beak hits a tree, its head is subject to 1,000 times the force of gravity.



How about walking?

There is more CO₂ (carbon dioxide) in the atmosphere today than at any point in the last 800,000 years.

Bazaar

NVIDIA GeForce GTX TITAN X

NVIDIA's latest and Maxwell's greatest has finally arrived

The GeForce GTX TITAN X is part of the elite TITAN family. For those who're looking at a TITAN for the first time, these cards make the most of each generation of NVIDIA's graphics architecture. The TITAN X is the fourth card after the TITAN, TITAN Black and the TITAN Z. The key difference is that the previous three cards were all based off the NVIDIA Kepler architecture while the TITAN X is based on Maxwell. And those who were hoping for the Maxwell pricing strategy to continue will be disappointed to know that the TITAN X was launched for \$999 which is what NVIDIA flagships were priced at till the Kepler architecture was ruling the roost.

What's new?

We first heard of the TITAN X back when it was showcased at GDC a few months back. The specifications weren't revealed back then but there were rumours of the card sporting a massive chunk of RAM and surely enough, the TITAN X does have 12GB of VRAM.

Like every TITAN that has come before it, the TITAN X is the most that you can expect from the Maxwell architecture. There are rumours of a 980 Ti but it'll be slotted somewhere between the GTX 980 and the TITAN X. Looking at the specifications, we can see that the TITAN X has 8GB VRAM in excess of the 980. But at the core, it has 8 SMs more which translates to 1024 CUDA cores, 64 TMUs and 32 ROPs. Essentially, the TITAN X has 50 per cent more resources than the GTX 980 which is quite promising since 50 per cent more resources should provide a massive boost in performance.

Also, the stock card that we tested was clocked at 1000 MHz and the VRAM was clocked at 1753 MHz. The base clock is a step down from the GTX 980 but way higher than its predecessor, i.e. the TITAN. However, the extra resources will more than make up for it. The memory clock may be the same as that of the GTX 980 but the TITAN X now has a 384-bit bus width that will provide the much needed head room to access all of the 12GB that comes with this new GPU. With such increased processing power, one can also expect a bump when it comes to the TDP and rightly so, the TITAN X is rated at 250W which is the same as the TITAN Z and the TITAN.

The previous members of the TITAN family also focussed on compute performance but the TITAN X seems to have its focus down to gaming perfor-



Price
₹85,099

mance only. The extra SMs do produce an increment in compute performance but it's nowhere close to the bump that the original TITAN generated compared to the GTX 780. So when you are looking for compute/price then the TITAN takes the back seat compared to the R9 295X2.

Build

Other than the all black look there seems to be virtually no difference at all so we really can't say any more for the TITAN X than what was said for the TITAN and the GTX 980. The black-plate we saw on the GTX 980 is no more. It has two power connectors (8-pin + 6-pin) and support for 4-way SLI. The external dimensions remain the same. Even the cooling setup is the exact same, so basically NVIDIA has let the design be as it was.

Performance

We ran the benchmarks on Full HD, 2K and 4K resolution and repeated the entire process after overclocking the card. An increment of 236MHz over stock was what we managed to achieve on air cooling and the results were pretty great. At stock the card showed a 20 per cent improvement in FPS scores for Bioshock when compared to the GTX 980. This massive bump was indeed a sight for sore eyes given the dismal improvement that Maxwell brought to the table with the 960, 970 and the 980. For example, Bioshock Infinite ran at 124 FPS on the GTX 980 and 149 FPS on the TITAN X. After overclocking, this rose to 160 FPS. The increment was seen across the board with all other games as well.

On the other hand, compute performance showed an increment of 29.5 per cent increment in LuxMark. Whatever, they've done with this card, we'd like to see more of it.

Mithun Mohandas



Performance..... 80
Value 75
Build 60

Specifications

Chipset: GM200; **Base clock:** 1000 MHz; **Memory clock:** 1753 MHz; **Stream processors:** 3072; **Texture Units:** 192; **ROPs:** 96; **Manufacturing process:** 28nm, 5120 x 3200 digital resolution support, 12 GB Memory; **DirectX support:** 12; **OpenGL support:** 4.5; **Power Connectors:** 8Pin + 6Pin; **TDP:** 250W; **Warranty:** 3 years

Contact

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